



Ing. Ákos Kappel

Full-Stack Software Engineer

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Summary

Full-Stack Software Engineer building web and mobile products end to end: React and TypeScript frontends, Kotlin Spring Boot and Elixir Phoenix backends, and Kotlin Multiplatform apps for Android and iOS. I focus on fast, accessible interfaces, type-safe code and architectures that are easy to extend, from finding N+1 queries with flame graphs to polished animations. I pick up new stacks quickly when a project needs them (Elixir, Ruby, KMP), and lately I work more and more with AI.

Technical Skills

- **Daily:** TypeScript, React, Kotlin, Spring Boot, TanStack (Router, Query, Form, Table, Virtual), Tailwind CSS, PostgreSQL, Git
- **Mobile:** Kotlin Multiplatform, Compose Multiplatform, Ktor, Koin
- **Worked with:** Python, Elixir, Phoenix LiveView, Node.js, Express, PHP, Laravel, Nette, Next.js, Vue, Nuxt, MongoDB, MSSQL, Zod, Vitest, Jest, Playwright, Cypress, GitLab CI, Docker Compose, Airflow, Ruby, Linux
- **AI and ML:** PyTorch, TensorFlow, OpenCV, OpenAI API; AI-assisted development with Claude Code, agent skills and MCP servers
- **Familiar:** Java, C#, Django, FastAPI, Redis, Bootstrap, Figma, Jira

Work Experience

Full-Stack Software Engineer, IGT Systems (formerly IGT Consulting)

February 2025 - now
Bratislava, Hybrid

- **Core3** (business platform: project management, attendance, CRM, HR, ticketing; team of up to 8): owned the Project Management module, including kanban with drag and drop and per-status rules, task permissions, and Scrum and Waterfall methodologies. Made slow pages fast after user complaints: profiled with flame graphs, removed N+1 queries in PostgreSQL and MongoDB, added caching with invalidation, async streams with infinite scroll, ordered PubSub updates, loading skeletons and optimistic UI. *Elixir, Phoenix LiveView, PostgreSQL, MongoDB, Cypress*
- **WYDO** (sports activity search, team of 3): started the Kotlin Multiplatform app for Android and iOS (map with clustering, search and filters, groups, chat over WebSockets, 4 languages), then the promo site, then moved the web from KMP to a standalone React app with virtualized feeds, push notifications, passkey login, per-route SEO and PWA support. Backend features in Spring Boot, including full WebAuthn passkey support. *Kotlin, Compose Multiplatform, React, TanStack, Spring Boot, Vitest, Playwright*
- **CleanEEG** (EEG cleaning service, team of 2): built the frontend from scratch, including a fast WebGL EEG viewer that compares raw and cleaned signals with client-side filters, keyboard controls and multi-channel layout, plus signup, billing and the file dashboard. *React, TypeScript, TanStack, Plotly*
- **Internal analytics dashboard** for a large enterprise client (team of 3): built the frontend for monitoring data pipelines, designed so new sections are added as routes from reusable parts, such as a generic typed data grid with nested rows and table state kept in the URL. *React, TypeScript, TanStack Query, Zod*

Full-Stack Developer, Švarba s.r.o.

June 2024 - January 2025
Nové Zámky, Remote

- Back-office system for an international direct-sales company: a new React web app that joined and replaced older Laravel and Nette monoliths.
- Designed and built the TypeScript REST API in Express (MSSQL with stored pro-

cedures, JWT auth, generated OpenAPI docs, cron jobs, WebSocket notifications over Redis, automatic translations with the OpenAI API).

- Wrote 200+ unit and integration tests (Jest, supertest); maintained and extended the legacy PHP system (reports, grids, T-SQL).
- *TypeScript, Node.js, Express, React, PHP, Nette, MSSQL, Redis*

Junior Python Developer, SoftPoint s.r.o.

October 2023 - May 2024
Bratislava, Hybrid

- Built ETL pipelines in Airflow that import product, price and stock data from supplier websites, APIs and XML, XLSX and FTP feeds into the data warehouse for a large e-commerce retailer.
- Automated supplier B2B portals with Selenium scraping and headless-browser ordering scripts, which I wrote in Ruby after learning it for the task.
- *Python, Airflow, pandas, Selenium, Ruby, PostgreSQL, FastAPI*

Projects

Self-hosted home server

2026 - now

- Run my portfolio demos on my own server: one Docker Compose project per app, published through Tailscale Funnel sidecars, nginx with rate limiting, nightly demo resets, and central traffic monitoring and attack detection (GoAccess, CrowdSec). Static apps deploy to GitHub Pages with GitHub Actions.

Modern Fashion Store | [GitHub](#)

2025, upgraded 2026

- Inventory and sales dashboard for a clothing company with live updates between users. Upgraded in 2026 to Phoenix 1.8, LiveView 1.2 and Tailwind 4, with a security review and a public demo that resets every night. *Elixir, Phoenix LiveView, PostgreSQL, Docker*

Glaucoma Segmentation (master's thesis) | [GitHub](#)

2022 - 2024

- Proposed and evaluated two neural network architectures (cascade and dual decoder) for optic disc and cup segmentation in fundus images. Best Dice score 96.8 % for the disc and 89.7 % for the cup, with the cup-to-disc ratio used for glaucoma screening. *Python, PyTorch, OpenCV, Albumentations*

Education

Ing., Intelligent Software Systems, Slovak University of Technology in Bratislava, FIIT

2022 - 2024

- *Thesis*: Neural network based semi-automatic segmentation methods to enhance the detection and monitoring of human eye diseases

Bc., Computer Science, Slovak University of Technology in Bratislava, FIIT

2019 - 2022

- *Thesis*: Problem generator for analytical geometry in the plane: conic sections

Languages

Slovak (native), **Hungarian** (native), **English** (C1), **German** (A2)

Interests

Self-hosting, Advent of Code, sports